Theme and chassis layout

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Make a dimensioned layout that supports the circuit and drivetrain.

SUCCESS CRITERIA - I can...

1. Label component locations and clearances.
2. Use dimensions to check that parts fit.
3. Explain a structural choice with evidence.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “chassis” and “clearance” to explain your model or result. Sentence starters are optional.

chassis: the frame that supports a vehicle’s components

clearance: space that lets neighboring parts avoid rubbing or colliding

constraint: a requirement or limit that the design must meet

01 Bring back an earlier idea

A 40-turn coil has flux per turn change by 0.006 Wb in 0.4 s.
Find average induced-emf magnitude.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: available length = chassis length - components - clearances.
Use measured dimensions for a top view. Keep axles parallel and controls accessible. Get approval before tools/cuts.

Worked reasoning: A 20 cm chassis reserves 6 cm for the battery, 5 cm for the motor and 2 cm total for end clearances. The remaining 7 cm must cover other components and gaps.

Your turn: Draw a dimensioned top view showing these regions.
Label any assumed dimensions rather than pretending they were measured.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Represent and interpret

Draw a dimensioned top view of your chassis.
Label motor, battery, switch, axle lines and wheel clearance.
Mark the intended forward direction.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Top view: the large rectangle is the chassis; small rectangles are wheels; the two crossbars are axles.

Marzano check - circle: ① ② ③

04 Solve and justify

A supplied 180 mm chassis needs two 15 mm end clearances and a 95 mm component zone.
Find remaining length.
Convert the chassis length to meters.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

Your theme adds a heavy decoration far to one side. **Predict** a possible effect and propose a way to test it without claiming the effect is certain.

Sentence starter: I predict ... . My model supports this because ... .

Marzano check - circle: 1 2 3

06 Choose a method

A 8 Ω resistor is in series with two parallel 12 Ω resistors across 28 V.
Find parallel equivalent, total resistance and supply current.

Word help: series = connected along one unbranched current path

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

07 Exit check

Explain why a top-view sketch alone may miss a wheel or gear clearance problem. Add one useful side-view dimension.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A 180 mm chassis needs two 15 mm clearances, a 60 mm battery and a 50 mm motor region.
Find remaining length. Then identify a side-view clearance that this calculation misses.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3